

RMO– 2016(3)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
 - All Questions carries equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Let ABC be a right angled triangle with $\angle B = 90^\circ$, let I be the incentre of ABC. Draw a line perpendicular to AI at I. Let it intersects the line CB at D. Prove that CI is perpendicular to AD and prove that $ID = \sqrt{b(b-a)}$ where $BC = a$, and $CA = b$.
 2. Let a, b, c be positive numbers such that $\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} = 1$, prove that $abc \leq \frac{1}{8}$.
 3. For any natural number n , expressed in base 10, let $S(n)$ denote the sum of all digits of n . Find all natural numbers n such that $n = 2S(n)^2$.
 4. Find the number of all 6-digits natural number having exactly three odd digits and three even digits.
 5. Let ABC be a triangle with centroid G. Let the circumcircle of triangle AGB intersect the line BC in X different from B; and the circumcircle of triangle AGC intersect the line BC in Y different from C. Prove that G is the centroid of the triangle AXY.
 6. Let $\langle a_1, a_2, a_3, \dots \rangle$ be a sequence of strictly increasing sequence of positive integers in an arithmetic progression. Prove that there is an infinite subsequence of the given sequence whose terms are in a geometric progression.

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